DIFFERENTIAL ITEM FUNCTIONING

Edited by

Paul W. Holland
Howard Wainer

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Educational Testing Service

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To Linda and Roberta
who make a big DIF in our lives

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Foreword: Differential Item Functioning (DIF): A Perspective From the Air Force Human Resources Laboratory

The Manpower and Personnel Division of the Air Force Human Resources Laboratory, part of the Air Force systems Command, was pleased to have co-sponsored a conference on DIF with the Educational Testing Service. We have long been partners in psychometric progress. The purpose of the American military is to deter aggression and defend against those who would deprive us of our rights. This is done in a manner consistent with fair treatment and equality. The twin ideas of human rights and fair treatment of humans have been advanced rapidly by the American military in this century. From the World War I committee of the American Psychological Association that met in Vineland, New Jersey to the 1948 presidential order integrating the armed services three decades later, to support for advanced psychometric models of fairness, to this conference, America's military research establishment has been in the forefront of development and fair application of advanced technology.

DIF is of concern to the Air Force and to all major test developers and users. The American military is in a unique position because we develop and use our own tests. This year, more than two million young men and women will be tested for enlistment qualification and additional hundreds of thousands of tests will be administered by the service for promotion and certification purposes. These tests will materially affect the lives of these military members and the security of our country.

The issue of test and selection fairness encompasses many concepts and models. Primary to all of them is DIF. If test items operate in a differential fashion, then the scores for different groups are *per se* not comparable. This cannot lead to equitable treatment, a declared goal of the American armed forces. For the Air Force, which produces both enlisted and officer tests, certain models

of DIF detection have become an integral part of the test production and evaluation procedure. Additionally, content and construct validation of our many tests benefit from DIF analyses. Finally, DIF models and knowledge point the way to fuller understanding of the characteristics that cause items to differ in item analytic difficulty and discrimination indices. In the future, DIF models will bring fundamental change to psychometrics.

This conference and the chapters of this book provide a solid foundation for the future development of knowledge and techniques to institutionalize fairness in our tests and our selection and classification systems. The knowledge provided by this conference has already been used in developing more equitable tests for the military and civilian community. Issues discussed at this conference and included in this volume will continue to advance research resulting in more equitable and better tests.

Malcolm James Ree

Preface

Test fairness is a moral imperative for both the makers and the users of tests. This has been a basic tenet for more than 30 centuries. Although formal testing was well under way during the Shang Dynasty (ca. 1760–1122 B.C.), by the time of the Han Dynasty (202 B.C.–A.D.220), written examination papers were rewritten by scribes. This rewriting was to ensure the anonymity of the examinees before they were graded, thus eliminating one possible barrier to the unbiased grading of the exams. Machine scoring of multiple-choice tests is a modern equivalent, but it is a trivial component of the modern armamentarium of techniques used routinely to ensure test fairness. Three of these, of relevance to the contents of this book, are:

1. Detailed reviews of test items by subject matter experts and members of the major subgroups in society (gender, ethnic, and linguistic) that, in prospect, will be represented in the examinee population.
2. Comparisons of the predictive validity of the test done separately for each of the major subgroups of examinees.
3. Extensive statistical analyses of the relative performance of major subgroups of examinees on individual test items.

This book is concerned primarily with number 3. With the exception of chapter 19, which is concerned with formal methods for implementing number 1, the focus of this book is very tightly focused on methods for detecting test items that function differently for different groups of examinees and on using this information to improve tests.

The history of this subject is littered with unfortunate, incomplete, and misleading terminology and vocabulary, which has led to ambiguous expositions and imprecise thinking. In this account, we have tried to maintain a consistent terminology and our authors have adhered to this quite well.

To begin, we reserve the ambiguous term *item bias* to refer to an informed judgment about an item that takes into account the purpose of the test, the relevant experiences of certain subgroups of examinees taking it, and statistical information about the item. In the main, we avoid using item bias in this book and refer, more neutrally and, to our way of thinking, more appropriately, to items that function differently in different groups of examinees. For short, we call this DIFferential Item Functioning, and, for shorter, simply DIF.

DIF is a relative term. An item may perform differently for one group of examinees relative to the way it performs for another group of examinees. The examinee group of interest is the FOCAL group, and the group to which its performance on the item is being compared is the REFERENCE group. In general, there will be several FOCAL/REFERENCE pairs of groups for with DIF analyses can be made.

In DIF analyses, items play a variety of roles. When an item is examined to ascertain the degree to which it exhibits DIF for a specific FOCAL/REFERENCE pair, it is the STUDIED ITEM for that analysis.

The genesis of this book reaches back at least to the mid-1980s, when the two of us often discussed the various approaches to measuring DIF. One focus of this discussion was our differing outlooks on what some call the *observed-score* versus the *true-score* approaches to the detection of DIF. Holland, then as now, firmly believes in the practical advantages and the conceptual clarity of the observed-score approaches, whereas Wainer was and remains convinced that a model-based approach is more principled, flexible, and powerful. In the intervening years both sets of beliefs have moderated somewhat. At that time, we agreed that we were making progress in understanding this general area of applied research and that a monograph providing both theoretical justification and practical advice would be possible within a few years. In consequence, we began to seek support for the research as well as for the conference that would lead to this book. Educational Testing Service (ETS) has provided very generous support to its staff for DIF research and for the preparation of this monograph. The Air Force Human Resources Laboratory (now Armstrong Laboratory) provided support for the conference that was held at the Henry Chauncy Conference Center at ETS in October 1989. Holland was able to finish his work on this book while a Fellow at the Center for Advanced Study in the Behavioral Sciences, with support from the Spencer Foundation as well as ETS.

This volume has benefited from the good thoughts, cooperation, and diligent efforts of a large number of people and it is our pleasure to have this opportunity to acknowledge them now.

First, we thank Gregory Anrig, the president of ETS, who has been a fervent

advocate of test fairness in general, and of this project in particular, since the very first days that he joined ETS. Nancy Cole participated both as an author in the book and as an interested senior manager at ETS whose background in the subject antedates our own. Our research vice president, Henry Braun, provided tireless support and has helped us in more ways than can be tersely listed here. Special thanks go to Malcolm Ree, of the Air Force's Armstrong Laboratory, whose concern with increasing the fairness and validity of testing has manifested itself in several fruitful collaborations between our two institutions.

The contributors to this volume have, and deserve, our warmest thanks. Their efforts to craft their words carefully to convey the results of extensive research has resulted, in our opinion, in a book of great value. We can only hope that they will have, by now, forgiven our occasionally impertinent requests for revisions and the seemingly endless and curious delays in the manuscript's final production.

Our good fortune to have Martha Byelich and Elizabeth Brophy as colleagues can hardly be fully recognized in a few sentences. Not only did they organize and run the conference that led to this book, but in good cheer they maintained timely control over numerous revisions of some manuscripts and, in a myriad of ways, provided the crucial help we needed to bring this book into being. We continue to wonder how they did it, but we are thankful that they did!

Paul W. Holland and Howard Wainer

INTRODUCTION AND BACKGROUND

Aristotle, in his *Metaphysics,* pointed out that "we understand those things best that we see grow from their very beginnings." We thus begin our examination of DIF with an over-the-shoulder look at the history of procedures that were developed to aid in the detection of malfunctioning items. In chapter 1, William Angoff traces the history of DIF methods from their heuristic beginnings to the more statistically rigorous methods currently available. He comments on the similarities and differences among these methods and on the language that has been developed to accurately describe exactly what it is that these methods detect. He also examines the very important set of arguments surrounding the necessity and importance of matching examinees in the two populations of interest prior to the calculation of any statistics characterizing the differential functioning of the item.

In chapter 2, Nancy Cole amplifies Angoff's concerns about the importance of a proper vocabulary to allow precision in the discussion of the technical aspects of test fairness. She then points out how an understanding of the social context in which discussions of test fairness take place are critical if one is to interpret those discussions properly.

1 Perspectives on Differential Item Functioning Methodology

William H. Angoff
Educational Testing Service

The studies of what has come to be known as *item bias* were apparently first undertaken in earnest in the 1960s. These studies were designed to develop methods for studying cultural differences and for investigating the assertion that the principal, if not the sole, reason for the great disparity in test performance between Black and Hispanic students and White students on tests of cognitive ability is that the tests contain items that are outside the realms of the minority cultures. The presumption was that these items deal with content that minority students have little occasion and less opportunity to learn. The more specific goal of these studies was, and continues to be, to identify any items that are biased against minority students and to remove them from the tests.

In parallel and contemporaneous with these studies have been the studies designed to investigate whether the tests as a whole, especially the tests used for selection, are biased against minorities, yielding lower scores for them than would be indicated by their standings on the criterion measures that these tests were intended to predict.

What set these two efforts apart, in addition to their targets of enquiry, and the different models and methods that were developed in their behalf, was that an obvious and natural external criterion existed for examining and evaluating bias in the selection test, whereas no such external criterion was thought to be readily available for evaluating bias in the items. As a consequence, investigators studying item bias resorted to the device of using the test as a whole, either directly or indirectly, as a surrogate criterion for matching the groups with respect to ability (or, to use the currently preferred term, *proficiency*). In both types of efforts it was recognized early that the study of bias would be falsely rendered if the criterion itself was inadequate in any way, most particularly if it was itself

biased. Under such circumstances it is quite likely that some biased items would be found to be unbiased and other, unbiased items found to be biased.

It may be useful at this point to consider the language that we ordinarily use in connection with studies of item and test bias—in particular, the use of the word *bias* itself. Several writers (e.g., Ironson, 1982; Linn & Drasgow, 1987; Linn, Levine, Hastings, & Wardrop, 1981; and Shepard, Camilli, & Averill, 1981) have defined item bias in very much the same way, essentially: An item is biased if equally able (or proficient) individuals, from different groups, do not have equal probabilities of answering the item correctly. (The same definition, in principle, applies to bias in the test.) It is of some interest that Shepard et al. also characterized bias as "a kind of invalidity that harms one group more than another" (p. 318). The first of these definitions, it is noted, refers to the simple observation of the difference in performance; the second contains the sense of an evaluation, over and above the difference itself, and refers to its (unfair) effect on a group. The dictionary defines bias as a divergence from some expression or indication of truth. Statistically, this translates into a tendency for an estimate to deviate in one direction or another from a true value. Socially, it means something akin to prejudice, an unreasoning inclination of temperament. When the concept first acquired popular psychometric currency in the 1960s, it was used in the sense used by Shepard et al.—that a possible reason, to be investigated, for the low test performance of minorities was that the items were capitalizing unfairly on knowledge and skills that were part of the White middle-class culture, but alien to the minority cultures. In an effort to investigate the validity of this hypothesis, methods were developed by the psychometricians, and studies undertaken, to determine how aberrant items, and therefore, by one definition, biased items, could be identified. But these were only statistical findings, subject to later interpretation and judgment. Some of these aberrant items may well have been biased, in the sense of being unfairly disadvantageous, against minorities. Some others, however, were judged to be quite fair, in the sense that they called on important educational outcomes, appropriate for all students, but unequally known and understood by all students. Still other items gave no clue as to the reason for their aberrancy.

In any case, it became clear very early that there was a semantic conflict here; the word *bias* was being used simultaneously, but quite understandably, for at least two entirely different meanings, social and statistical. The consequence was that unnecessary additional confusion was being introduced into an already confused political atmosphere in which bias was alleged to be the cause of the large disparity in scores. Some suggestions were made to use a term other than *bias* for the statistical observation, quite apart from its judgmental or interpretive meaning and use, and another term to describe the judgment and evaluation of bias in the social sense. Finally, the expression *differential item functioning* (DIF) came into use, referring to the simple observation that an item displays different statistical properties in different group settings (after controlling for differences

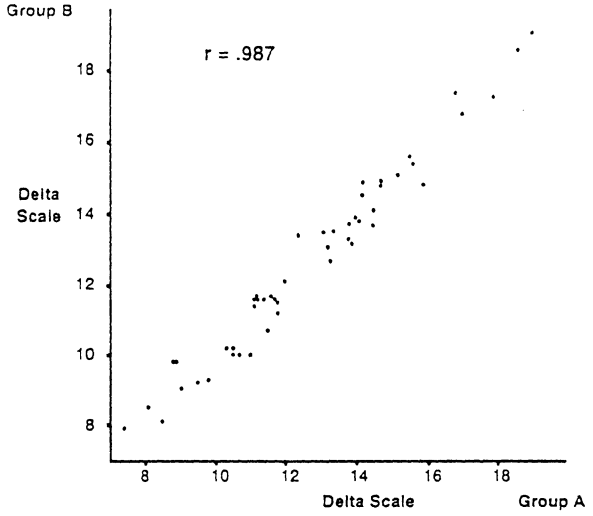
in the abilities of the groups, a topic discussed later). How the item was to be judged and used, that is, whether it was socially biased, and what disposition was to be made of it, was regarded as a quite separate matter. But semantic habits die hard. The word *bias* is particularly seductive; it is singular, brief, and concise, and lends itself comfortably to use as either noun, verb, or adjective. *Differential item functioning,* on the other hand, is long, unwieldy, and terminologically unadaptable. Nevertheless, it is the far preferable term for the objective observation, that is, without prejudgment, of a group disparity. Its freedom from evaluative overtones is worth preserving, permitting bias as a social issue to be examined quite separately.

METHODS FOR INVESTIGATING DIFFERENTIAL ITEM FUNCTIONING

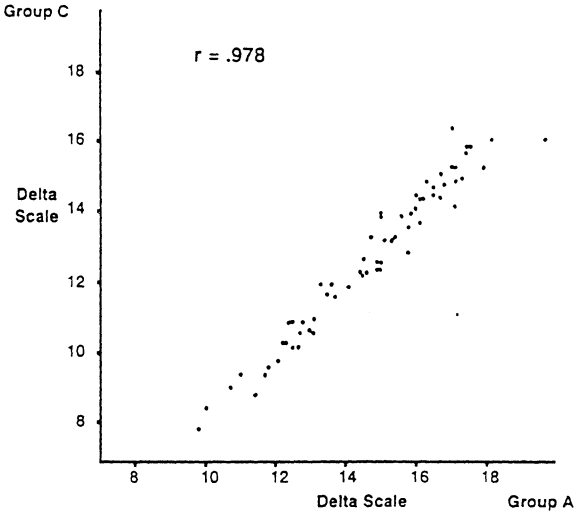
One of the first formal procedures undertaken to evaluate the extent of what was then, without the distinction just discussed, called item bias in a test was that by Cardall and Coffman (1964), in which they applied analysis of variance procedures to test the interaction of items with groups of Black and (mostly) White examinees who had taken the SAT in 1963. Essentially the same procedure was followed in connection with their studies by Cleary and Hilton (1968) and by Angoff and Sharon (1974), but the approach does not appear to have caught on very widely.

In 1972, Angoff offered a method for studying cultural differences, known variously as the *delta-plot* or *transformed item-difficulty* (TID) method. This method shortly became highly popular for reasons of its easy logic and practical simplicity, and for this reason it is described here in some detail. Although it was originally described by Thurstone (1925) in connection with his method of *absolute scaling,* the method was later applied by Tucker (1951) in a study of an academic ability scale of vocabulary items and by Gulliksen (1960) in a binational comparison of scale values of occupational prestige.

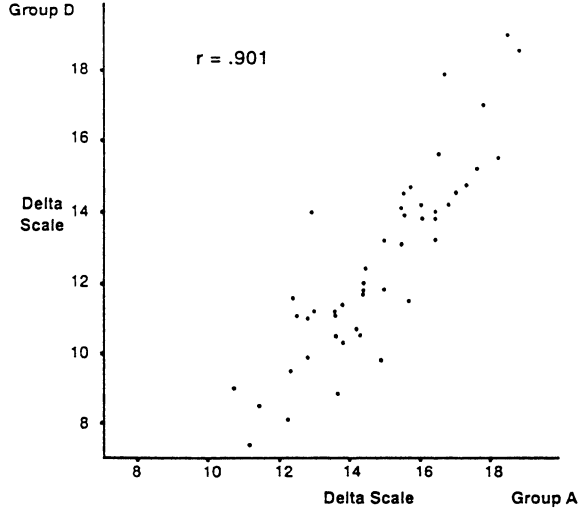
The delta-plot method calls for the calculation of item p-values for each of the two groups under consideration and for the conversion of each p-value to a normal deviate, usually expressed on a scale with a mean of 13 and standard deviation of 4. The pairs of normal deviates, one pair for each item, are then plotted on a bivariate graph with the two groups represented on the axes, each pair represented by a point on the graph. When the groups are of the same type and of the same level of proficiency, the plot of these points will ordinarily appear in the form of an ellipse extending from lower left to upper right, often representing a correlation of .98 or even higher, indicating that the rank order of difficulty of the items is essentially the same in the two groups. (See Fig. 1.1, panel a.) When the groups differ only in level of proficiency, the ellipse will be displaced vertically or horizontally, depending on which group is the more profi-



Panel A
Groups of the same type, equally proficient



Panel B
Groups of the same type, not equally proficient



Panel C
Groups of different types, not equally proficient

FIG. 1.1. Delta plots for different pairs of groups.

cient. (See Fig. 1.1, panel b.) However, when the groups are drawn from different types of populations, the points will be dispersed in the off-diagonal direction and the correlation represented by the points will be lower than previously indicated. The items falling at some distance from the plot of points, as measured by the distance of the item's bivariate point from the principal axis of the plot, may be regarded as contributing to the item × group interaction (Angoff & Ford, 1973). (See Fig. 1.1, panel c, in which the groups are also different in levels of proficiency.) These are the items that are especially more difficult for one group than for the other, *relative to* the other items, and are ordinarily taken to be characterized by DIF. The point is worth making here that the relative proficiencies of the two groups is of no particular significance in applying the method. What is important is the position of the points in the graph relative to one another.

A significant development, punctuating a major breakthrough in the history of test theory, was reached in the publication of F. M. Lord's dissertation (Lord, 1952) and in the Lord and Novick (1968) text, in which Lord explicated the *item response theory* (*IRT*) model. Within a short time thereafter it became clear that this model could be profitably applied in the study of differential item functioning. As is by now well known, the basis of the theory lay in the item response function, the S-shaped trace of the proportion of individuals at the same ability level who answer a given item correctly. Under the assumption that the ability under consideration is unidimensional and that the item measures the same ability, the trace is unique under the conditions of a particular model; except for random variations the same curve is found, irrespective of the nature of the group for whom the function is plotted. The curve is often defined by three parameters: a, which is proportional to the slope of the curve at the point of inflection and represents the discriminating power of the item; b, the proficiency level, θ, at the point of inflection, representing the difficulty of the item; and c, the lower asymptote—alternatively called the *pseudo-guessing parameter*—corresponding to the probability that a person completely lacking in proficiency will answer the item correctly.

Because of the unique nature of the item response curve (under the assumptions already stated), the finding that the response curve is not in fact the same for two groups is evidence that the assumptions are not satisfied for one or both groups, and statistical tests are available for testing the significance of the differences between the curves with respect to the three parameters (Thissen, Steinberg, & Wainer, 1988). There are also tests for examining the areas between the curves—that is, the degree to which the curves fail to coincide.

Unlike other methods of studying item bias, the use of the three-parameter item response model is a comprehensive one, taking into account not only differences between the groups with respect to item difficulty, but also differences with respect to discriminating power and differences with respect to the

pseudo-guessing parameter. Other methods either ignore differences among items with respect to discriminating power and guessing or assume that they do not exist. The fact that such differences among items, especially in discriminating power, do exist has made the IRT model the criterion method to use for studying item bias and for identifying biased items. Its use as a standard procedure has been inhibited, however, by the large amounts of data required to yield stable findings, its attendant cost, and, more than likely, the complexity of the model itself and the frequent failure of operational tests to satisfy the assumptions of the model. Other methods, like the delta-plot method, are far less expensive, require less data, and are intuitively easier to understand and apply. However, as is described later, the delta method is now considered to be technically flawed.

The Rasch model, a one-parameter item response theory (Rasch, 1960; Wright & Panchapakesan, 1969), is one that assumes that all items are equally discriminating and that items are not answered correctly by guessing. In effect, its use for examining items for bias is limited to an examination of differences in item difficulty (after conditioning on ability). Because it assumes that there are no differences in the *a*- or *c*-parameters, any such real differences are therefore not detectable by means of the Rasch model. Even more serious is the fact that any real differences in the *a*- or *c*-parameters are likely to result in artifactual DIF values.

The foregoing is similarly true of the delta-plot (TID) method. It has been pointed out by Cole (1978), Cole and Moss (1989), Hunter (1975), Shepard (1981), and Lord (1977), and affirmed by Angoff (1982), that unless all the items in a delta plot have the same discriminating power, the method may yield misleading results, especially when the groups studied earn mean scores at widely different ability levels. The delta, it is recalled, is taken from the proportion of individuals in the two groups answering the item correctly. These proportion-correct values may be found on the item response curve corresponding to the mean ability score of the group. If one compares the difference in proportion-correct figures for the two groups under consideration for two items with different discriminating power, one sees that the difference is smaller for the less discriminating item than for the more discriminating item. (See Fig. 1.2.) This being the case, it is likely that at least some of the items identified as unbiased by the delta-plot procedure may in fact be found to be biased, if appropriate account were taken of their relatively low discriminating power. The reverse may similarly be true: Items identified as biased by the delta-plot method may in fact not be biased, but only give a biased appearance because they are more discriminating. Both types of errors would be unfortunate and costly—the first, because some biased items would go undetected; the second, because it could lead to the ultimate removal of the most useful items in the pool.

Lord (1977) was critical of proportion-correct figures for several reasons,

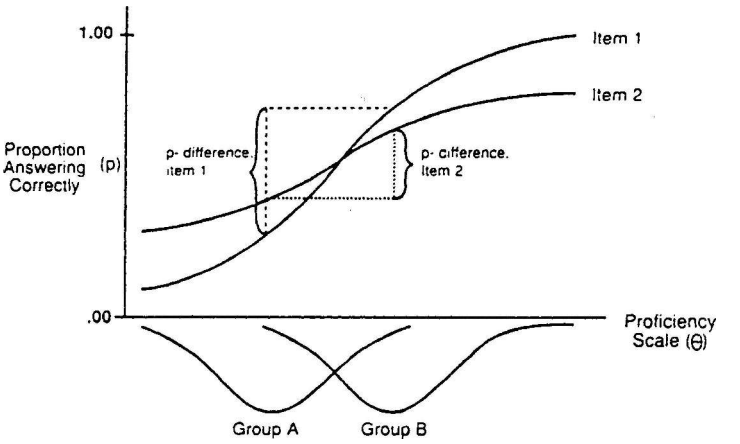


FIG. 1.2. Item response curves for two items of different discriminating power, taken by two groups of different levels of proficiency.

among them that they are group-dependent and therefore confounded values, sometimes leading to curious anomalies. For example, if we have two items with different *a*-parameters and two groups of different average ability, item 1 may turn out to be more difficult than item 2 for Group A, but less difficult than item 2 for Group B (Lord, 1980, p. 35). Unlike proportion-correct figures, the item-difficulty parameter, derived from the three-parameter IRT model, is invariant with respect to group.

Shepard et al. (1981) found that the one-parameter (Rasch) index of DIF yields a correlation higher than .99 with the TID index of DIF. This should not be overly surprising. It can be shown that the *b*-value derived from the one-parameter model is directly proportional to the delta. This would imply that whatever errors of interpretation are attributed to the delta-plot procedure are shared by the Rasch model.

In 1979, Scheuneman published a method similar (but not identical) to chi-square for evaluating DIF, but this method was subsequently criticized by Baker (1981) for yielding values irrelevantly affected by sample size and with an unknown sampling distribution—in effect, not a chi-square procedure at all. A corrected statistic, sometimes called the "full" chi-square, appears in Bishop, Fienberg, and Holland (1975). The full chi-square procedure calls for dividing the groups into intervals of total score (Scheuneman [1979] suggested 4 or 5 intervals) and forming a 2×2 table for each interval, showing pass-fail on one axis and the two groups on the other axis, as shown here for the *i*th interval. A chi-square is calculated for each of the tables (intervals) and summed. The number of degrees of freedom is the number of such intervals.

		Performance on Item i: 1	0	
Tested Group	r	a_i	b_i	$N_{r_i} = a_i + b_i$
	f	c_i	d_i	$N_{f_i} = c_i + d_i$
		$N_{1_i} = a_i + c_i$	$N_{0_i} = b_i + d_i$	$N_i = a_i + b_i + c_i + d_i$

Nine years after Scheuneman's (1979) development, Holland and Thayer (1988) described a procedure for investigating DIF that made use of a technique designed by Mantel and Haenszel (1959) in their retrospective studies of disease. The Mantel–Haenszel procedure may be described as follows: Assume, as in the full chi-square method, that the 2 × 2 table shown here describes the frequencies of correct (1) and incorrect (0) performance of the focal (f) and reference (r) groups (where *focal* refers to the particular group of interest—Blacks, Hispanics, Asians, Native Americans, women, etc.—and *reference* refers to the group with whom the focal group is to be compared—Whites, males, etc.), all of whom score in the same score interval on the relevant section of the test, usually the section of the test that contains the item. In this operational sense, the focal and reference groups are matched on the ability most nearly relevant to the ability measured by the item.

The Mantel–Haenszel index is calculated at each score interval (i) of the section of the test of which the item is a part:

$$\alpha_i = \frac{p_{r_i}}{q_{r_i}} \Big/ \frac{p_{f_i}}{q_{f_i}} = \frac{\frac{a_i}{a_i + b_i}}{\frac{b_i}{a_i + b_i}} \Big/ \frac{\frac{c_i}{c_i + d_i}}{\frac{d_i}{c_i + d_i}} = \frac{a_i}{b_i} \Big/ \frac{c_i}{d_i} = \frac{a_i d_i}{b_i c_i}, \tag{1}$$

where p_{r_i} is the proportion of the reference group in score interval i who answered the item correctly, and $q_{r_i} = 1 - p_{r_i}$. Similarly, p_{f_i} is the proportion of the focal group who answered the item correctly, and $q_{f_i} = 1 - p_{f_i}$. Thus, α_i is the ratio of the odds (p/q) that the reference group students have answered the item correctly to the odds that the focal group students have answered the item correctly. If there is no difference in the performance of the two groups on this item within this score interval, then α_i will be equal to 1. If, however, the two groups function differently—if, for example, in this score interval the focal group performs better on the item than the reference group, then $\alpha_i < 1$. If, on the other hand, the reference group performs better than the focal group, $\alpha_i > 1$.

The Mantel-Haenszel procedure estimates a common odds ratio across all matched categories. The form of its index is given as follows:

$$\hat{\alpha}_{MH} = \frac{\sum_i p_{r_i} q_{f_i} N_{r_i} N_{f_i}/N_i}{\sum_i q_{r_i} p_{f_i} N_{r_i} N_{f_i}/N_i} = \frac{\sum_i a_i d_i/N_i}{\sum_i b_i c_i/N_i}, \quad (2)$$

which is the average factor by which the odds that a member of the reference group responds correctly to the item exceeds the odds that a member of the focal group responds correctly to the item. It is observed that the index is weighted by the number of cases in the interval; also, that the interval in which the numbers of cases in the two groups are more nearly equal receives the heavier weight.

For the sake of convenience, $\hat{\alpha}_{MH}$ is transformed to another scale, yielding an index that is referred to as *MH D-DIF,* by means of the conversion, MH D-DIF $= -2.35ln(\hat{\alpha}_{MH})$. This transformation centers the index about the value 0 (which corresponds to the absence of differential item functioning), and puts it on a scale roughly comparable to the ETS delta scale of item difficulty and *reverses* the index so that positive values of MH D-DIF indicate that the item favors the focal group; negative values indicate that the item favors the reference group and disfavors the focal group.

Holland and Thayer (1988) suggested that the Mantel–Haenszel procedure, which provides both a significance test and a measure of the effect size ($\hat{\alpha}_{MH}$, or MH D-DIF), enjoys statistical properties to be preferred over the full chi-square methods, such as those offered by Bishop et al. (1975) and by Marascuilo and Slaughter (1981), and point out, as did Angoff (1982), that statistical tests will always show significance if the sample sizes are large enough. The measure of effect size will serve to temper judgments influenced by significant results when it is seen that the effect size is actually small.

Dorans and Kulick (1986) offered a method of identifying DIF, called the *standardization method,* which makes use of the same information as does the Mantel–Haenszel procedure and is quite similar to it in other respects; indeed, the correlation between the two DIF indices, when expressed on the same scale, is .99 (Wright, 1987a). The principal differences are that, first, at each interval on the dimension of the matching variable the standardization method considers the difference in *p*-values for the focal and reference group (which Dorans & Kulick, 1986, call the *base group*); second, it weights these differences in terms of a specially identified *standardization group,* typically the focal group. Thus,

$$D_{STD} = \sum_i k_i(p_{f_i} - p_{b_i}) \Big/ \sum_i k_i, \quad (3)$$

where i refers to the score interval of the matching variable and $k_i/\sum_i k_i$ is the weighting factor at score level i supplied by the standardization group and used to weight differences in performance between the focal group (p_{f_i}) and the base group (p_{b_i}). Typically, the focal group has served as the standardization group.

An important advantage of the standardization method, not shared by the

Mantel–Haenszel method except by approximation, is that it yields a direct estimate of the weighted difference in *p*-values between the focal and reference (base) groups. This difference can just as easily be expressed, if one wishes, in terms of deltas, the normalized equivalents of *p*-values expressed on a 13-4 scale (as described previously).

As has been clear in the previous discussions, the methods of identifying DIF typically focus on the correct option. But the incorrect options (distractors), and the embedded and trailing omits, also exert a major effect on the difficulty of the item, and it is likely that the source of the DIF will occasionally be found there. Dorans, Schmitt, and Bleistein (1988) and Green, Crone, and Folk (1989) recently did some developmental work on a DIF model for other-than-correct responses, and further work in this area is expected.

Like the delta method and the full chi-square method, but unlike the IRT method, the Mantel–Haenszel and standardization methods provide indices of differential item difficulty, but not of differential item discrimination. Indeed, it is possible for an item with the same difficulty parameter in the two groups but with different slope parameters to yield a DIF index of zero when analyzed by all but the IRT method. It would therefore be advisable to plot the component chi-square values, or MH indices, or standardization indices score interval by score interval, as was done by Dorans and Kulick (1986), as well as to calculate a weighted index across the entire range of score intervals. An upward or downward trend in this plot would warrant the conclusion that there is a difference between the two groups with respect to the item's discriminating power.

OTHER CONSIDERATIONS

Characteristics Common to All Methods

It is significant that all the methods available to identify DIF are designed to match the groups, either directly or indirectly, on the proficiency measured by the items. Even the delta-plot method, which does not normally use an explicit matching variable, matches the groups nevertheless, in the sense that group differences are observable, but for all practical purposes in using the method, they are ignored; what is of interest in examining the results is the off-diagonal dispersion, the appearance of anomalous, that is, aberrant, item points relative to the other item points in the swarm—in effect, the outlier items that contribute to the item × group interaction.

The sense of conditioning on a criterion variable (usually the total score on the section that contains the item or, in the case of the delta method, the collection of other items in the section) is common to nearly all methods of detecting DIF, including a method described by Stricker (1982), which calls for the correlation between item and group membership, with true scores on the total (criterion)

score partialled out. The general notion of conditioning on a criterion variable in the process of examining group differences is also illustrated in the Cole (1973) model of test bias, in which, for a given score on the criterion, the average test scores of the two groups being compared are examined.

When explicit matching is carried out on an observed score, as it is for the chi-square, Mantel–Haenszel, and standardization methods, it appears that the matching variable should be stratified as finely as possible, consistent with the amount of data on hand. To use broad strata would allow group differences within the strata to dilute the effect of the matching.

It is also significant that all the methods and techniques that have been developed to identify DIF in the items assume that the group of items, or the test that contains the items, is homogeneous and unidimensional. This assumption is explicitly true of the IRT models and, as just indicated, at least implicitly true of the delta-plot method and the chi-square, Mantel–Haenszel, and standardization methods. To test this assumption conceptually, one might imagine an analysis involving matched groups of men and women, of a test containing both verbal and mathematical items. The results of such an analysis would undoubtedly show a strong tendency for the verbal items to favor women and for the mathematical items to favor men, thus providing evidence that verbal and mathematical items do indeed represent different domains.

The Golden Rule Settlement

In recent years a question has arisen at the level of social policy in connection with the settlement between the Golden Rule Insurance Company on the one hand and the Illinois Department of Insurance and Educational Testing Service (ETS) on the other (Faggen, 1987), regarding the way in which items would be considered for inclusion in the Illinois insurance licensing examinations. It was agreed as part of the settlement—there were other conditions as well—that a raw difference, favoring White applicants over Black applicants, of .15 or more in an item's p-values was to be taken as evidence that the item is to be considered biased in the social sense, that is, unfair to the lower-performing group, and identified as an item not normally to be included in the test. The virtually unanimous view of the psychometric community (Bond, 1987, referred to it as "axiomatic"), however, is that a simple raw difference in favor of the White group with respect to their average success rate on a test item is, by itself, insufficient reason for concluding that the item is biased against the Black group. The same consideration applies to mean scores on a test.

There are two lines of evidence to support this view. First is that just as we have reason to expect individuals to differ consistently among one another in test performance, so do we also have reason to expect that entire groups will differ consistently among one another in test performance. This is not to say that such differences are innate and unchanging, or that they in any sense reflect a funda-

mental difference in ability. Indeed, we know that individual and group differences on cognitive tests are in very large part brought about by differences in the nature, quality, and amount of education, both in school and out of school, to which people are exposed. This is particularly true of Black people in our society. We know that Blacks in America had been for many years excluded, both by law and by custom, from the normal educational stream. It should therefore come as no surprise to us that they typically score lower today, on the average, than White groups. Clearly, there is bias here, but just as clearly, the bias resides in the treatment that has been historically accorded these groups and which now reflects itself in general academic performance, including, specifically, performance on the tests. To deny the reality of the differences in test performance—at least, the likelihood that the differences are real—is to deny that there is a real difference in the academic knowledge and skills that the tests are designed to measure and to deny that some minority groups in this society are indeed educationally disadvantaged. It would say, in effect, that the educational opportunities afforded minority groups in this society are, and have consistently been, equivalent in quality to those afforded the mainstream groups. This is clearly not the case.

As already discussed, the task of the various models of DIF is to identify those items whose behavior is anomalous, inconsistent with the message of the test as a whole. All the models of DIF recognize, either directly or indirectly, the strong likelihood that some component of the observed score difference between the groups under study—including White and Black groups—is a real difference. It is in this sense that they all aim to separate bias in the item, if it exists, from the bias in society, as discussed previously, that reflects itself in group differences in school performance and in test performance as well.

Quite apart from the more general logic that supports the examination of group differences after first matching for ability, there is a specific psychometric aspect to this logic as well. Linn and Drasgow (1987) considered the position embodied in the Golden Rule settlement and have made it clear that, among other things, the settlement undermines the cause of good measurement: In calling for the considered exclusion of items that show relatively large disparities (p-value differences greater than .15) between White and Black examinees, the settlement works toward the avoidance of the most discriminating items, those that distinguish between high- and low-performing groups, in favor of the less discriminating items. The inevitable outcome of such an effort is that the resulting test will necessarily be a less reliable, less psychometrically discriminating, and ultimately less valid instrument.

It was asserted earlier that there were two arguments supporting the view that a raw difference between the two groups in item performance is not by itself sufficient reason for concluding that the item is biased against the lower-performing group. One reason, already given, is that a large difference is to be expected in view of the history of educational experiences of the lower perform-

ing Black group. The record of poverty today and the educational ills that typically accompany poverty today—low rates of school attendance and high rates of dropout, poor academic performance, low representations in national competitions (Merit scholarships, Westinghouse awards) and recognitions (e.g., Study of Mathematically Precocious Youth)—provides convincing evidence that the principal motivations of many young Black people in our society today do not run along academic and intellectual lines. The implication one would be forced to draw from the Golden Rule settlement, on the other hand, is that the entire disparity between White and Black test performance is accounted for by bias in the test, that in point of fact there is no real difference in the knowledge and skills of the two groups and that Black students are in fact not educationally disadvantaged.

The second argument is one that was alluded to early in this chapter, that statistical evidence of a large difference in the groups' performance, even evidence of DIF itself, is not by itself entirely persuasive in deciding whether or not to remove an item from the test; there may be good reason for retaining the item in spite of the statistical findings. For example, Doolittle and Cleary (1987) found that female students do worse than male students of the same general mathematical ability on geometry and mathematical reasoning items. If, then, one were to follow the results of the analysis mechanically, one would remove such items from the test. But this obviously makes little sense. The understanding of geometry and the ability to reason mathematically represent legitimate and necessary goals of mathematics education, and students, whoever they are, should be expected to cope successfully with problems of this sort. To remove such items would not only render the test incomplete, it would be disadvantageous to the students who failed them, for it would serve to conceal the particular source of their mathematics deficiency and perpetuate it. Shepard (1987a) expressed a similar view: "[T]o the extent that members of one group do more poorly on a subtest of items that are a *legitimate part of the content domain,* we would be reluctant to call the discrepancy evidence of *bias*" (p. 186, italics in original).

The process of matching the two groups for ability, although certainly a preferred procedure, nevertheless does raise, in the public's mind, some interesting questions of its own. For example, when there already exists a marked difference between the Black and White groups, what does it mean to "match"? If we select from the Black and White groups the same number (or proportion) of individuals, score level by score level, we have indeed conditioned effectively on ability (proficiency), as measured by the total test score. At the same time, however, have we not produced for study a group of Blacks who are unrepresentative of Blacks generally and a group of Whites who are similarly unrepresentative of Whites generally? And is it now possible that, when we have conducted the DIF analysis, we have in fact assembled a set of data for an irrelevant comparison, between groups who are not the original groups of interest? Does

the analysis now have the intended value? In reply to this concern we can say that we have extracted the ability differences between the groups and have laid bare for comparison the remaining differences, those having to do with the students' color, culture, and group identity. In the last analysis, this is the essence of the comparison we wish to make.

Characteristics of the Matching Criterion

There is unanimous agreement that the matching criterion should be valid, reliable, and fair to be acceptable. It is also important that it be so perceived. For if the criterion is itself biased to some degree, then the application of a DIF analysis will certainly be flawed; further, if bias is pervasive in the criterion, then any attempt to identify bias in its component items will inevitably fail. This is the weakness in a circular procedure, in which the "criterion of truth" is not only a test—and therefore already suspect in the minds of some—but is also the test that comprises the very items whose possible bias we seek to determine. On the other hand, the circularity cannot be avoided; the test itself is the most relevant and reliable criterion available to help answer the important questions relating to the validity of the test and the inferences one is to draw from it. And from a statistical point of view it appears that if the studied item is not included in the matching criterion, then the Mantel–Haenszel procedure will not be optimal. Holland and Thayer (1988, p. 141) suggested a two-step procedure for examining items for DIF: "Step 1: Refine the matching criterion by eliminating items based on a preliminary DIF . . . analysis. Step 2: Use as the matching criterion the total score on all items left in the refined criterion plus the studied item—even if it is then omitted from the criterion of all other items when they are studied in turned (*sic*)." This iterative procedure, which successively refines and purifies the criterion over the course of several analyses, should satisfy the concern that the procedure is circular and therefore suspect.

Interpretation of DIF

It is to be emphasized that finding DIF in an item does not necessarily imply that the item is biased, that is, unfair to one of the groups. Considerations of unfairness would imply normative (competitive) comparisons with other individuals, in instances where, for example, the examinee earns what is considered to be a reward for a relatively high score or a demerit for a relatively low score. But there are also instances when such judgments and actions are not the issue, when, for example, two different language or cultural groups are being compared on a common measure as part of a research effort in linguistic comparisons (e.g., Alderman & Holland, 1981; Angoff & Sharon, 1974). Nor is it necessarily the case that unfairness exists in the item itself; often an item showing even substantial DIF may be judged to be quite fair, depending on its purpose. Ironson (1982)

cited an example of an item asking for the temperature required for baking a cake. The item, even if statistically biased against males, may be quite appropriate if it is used in a test to select cooks and bakers.

The question then arises of how, in general, items that function differentially are to be regarded and interpreted. There appears to be general agreement that, cast most easily in the sense of the IRT model, items that do not yield the same item response function for two or more groups are violating one of the fundamental assumptions of item response theory, assumptions also implicit in the other models of DIF, namely that the item and the test in which the item is contained are measuring the same unidimensional trait (e.g., Linn et al., 1981; Shepard, 1987a). A large DIF value would suggest that the item is measuring an additional construct in one of the groups that may not be relevant to the intended construct of the test. As a consequence, the conclusion might be reached that the assumption of unidimensionality is not satisfied: The test is not unidimensional for at least one of the two groups, or does not measure the same dimension in the two groups. (This conclusion—which could be investigated in a study of the internal structure of the test, by, for example, factor analyzing the items of the test separately for the two groups (see Cole, 1981)—suggests that dimensionality is not a function of the test by itself, but a function of the test only in the context of a particular group of examinees. Moreover, the example, cited previously, of the differential difficulty in a mathematics test of mathematical reasoning items for one of the groups suggests that the general conclusion of multidimensionality might have to be further confirmed by an editorial examination of the item content.) In any case, inasmuch as the construct represented by the test may be different from one group to another, the inferences that may be drawn, and the judgments and actions taken on the basis of the scores, may similarly be different from one group to another. It is in this sense that the validity of the test as a basis for a particular inference is thrown into question, and it is in this sense that bias, as unfairness, is defined; that bias has to do with inferences, evaluations, and ethical considerations as they affect the test's use.

Use of Additional Matching Variables

It is useful to regard group differences in item (and test) performance as consisting, in theory, of two components: the "true," or unassailable, difference between the groups and an artifactual difference brought about by the use of inappropriate and irrelevant (DIF) items. These two effects, in sum, have sometimes been referred to as *impact*. The use of the total score criterion as a matching, or conditioning, variable, as a measure of the "true" proficiency of the groups, serves to separate out these two effects. But the separation can only be as effective as the relevance, or, operationally, the correlation, of the criterion variable with success on the item. Clearly, if the correlation is zero, the matching criterion is entirely ineffective in separating out these effects, and one has no

alternative but to refer to the group difference as undifferentiated impact. It is therefore important to choose a matching criterion that correlates best with the item; and if the test is truly unidimensional, the total score on the test is probably the one variable that correlates optimally with the item. But item-test correlations, even for reliable, well-constructed tests, do not average much above .50, suggesting that there is still a great deal of variance in the item left untapped by the total test score. This argues for the use of additional matching variables in an effort to increase the overall correlation with the item.

It has been reported by test developers that they are often confronted by DIF results that they cannot understand; and no amount of deliberation seems to help explain why some perfectly reasonable items have large DIF values. Several investigators (e.g., Cole, 1981; Linn, 1986; Plake, 1980; Shepard, 1981, 1982; Tittle, 1982) noted that the judgment of bias is generally unreliable. This is not surprising; the judgment of item difficulty by itself is not highly reliable (see, e.g., Thorndike, 1982), and the judgment that an item will or will not be differentially difficult is expectedly even less reliable. In one attempt to account for DIF results, Bleistein and Wright (1985) found, in a study of the performance of Asian-American students on mathematical word problems, that the lack of English fluency appeared to be the cause (or at least one of them) for their poor performance. In the case of mathematics achievement tests, Wise (1985) reported that she was able to account for a significant proportion (about seven eighths) of the mean male–female difference at the twelfth grade level by conditioning on the amount of math taken in high school and on ninth grade level math achievement. Further, it is not a difficult stretch to hypothesize that the solution of geometry problems calls on spatial ability (Maccoby & Jacklin, 1974). It would be useful to know to what extent women, who typically score lower than men on spatial tests, do worse on geometry problems for this reason. As yet another example, it is more than likely that controlling on one or more aspects of socioeconomic level might well explain some of the disparity in Black–White scores. In general, then, it is certainly possible that other variables, in addition to total score, may help to explain otherwise unexplainable results. It seems reasonable to test this possibility—to identify first those items with substantial DIF values, after controlling first on total score alone, then to add variables in order (either by matching or by forming a single weighted composite criterion) that, by hypothesis, show promise of explaining more of the variance in item behavior than initially found—to determine whether, in fact, there has been a reduction in the variance of DIF values as a result of using these additional matching variables. Any further operational decision to retain, revise, or discard such items would depend on the nature of the added variables and the judgment of the educational, psychological, and social mechanisms that caused the observed change in the DIF values. A principal caution to be observed in carrying out such studies is that the added matching variables are not of a sort that would interact with—that is, that they do not have different meanings for—the two groups, a

potential danger that may be particularly acute when we seek to condition on educational, occupational, or other social status variables.

The foregoing effort can also be accomplished by other means; for example, by studying groups that are matched in terms of age, grade level, and socioeconomic status, as described by Jensen (1980) and carried out by him and by Linn et al. (1981). Still other analyses could be conducted by studying differences between pseudogroups, that is, groups of the same race, but representing differences of a sort characterized by typical Black–White comparisons (Jensen, 1980).

At a more general level, a process of analysis such as this may also help us to clarify the notion of *differential opportunity to learn* in deciding on the appropriate disposition of an item. In discussions of this sort, it will be useful to avoid terms that, like this one, express an evaluation and judgment in advance of an examination of data, and to substitute a somewhat more parsimonious term, such as *differential educational exposure,* which avoids hypothesizing any mechanism that might have caused the disparity between the groups to occur.

Appropriate Criterion for the Judgment of Significance

On a previous occasion (Angoff, 1982), the suggestion was made that, when DIF is to be tested for significance, the use of random variation as a yardstick for comparison might be too stringent a source of variance for ordinary use. Instead, we might choose to identify and scrutinize only those items whose DIF values exceeded *acceptable* levels. Thus, for example, only values of DIF found in ethnic, racial, or sex studies that exceeded those found in studies of the test following its administration to two nonrandom groups—say, two groups of students choosing to take the test at different administrations, or two groups tested in geographically separated metropolitan centers—would be deemed significant, assuming that differences between such pairs of groups would be considered tolerable. The sense of this criterion is that if we are willing to tolerate a DIF of this size in other contexts, why not in the male–female (or Black–White, Hispanic–White, Asian–White, etc.) context?

Practical Considerations

One outcome of the DIF process that does promise to be troublesome has to do with the disposition of items with positive values of DIF, items that, after matching, appear to be *easier* for the focal group than for the reference group. It is recalled that the overall mean value of DIF, when the total test score is used as the matching variable, is defined as zero; therefore there will necessarily be a substantial proportion of positive values (or a small proportion of highly positive values). But the basic purpose of the DIF study is to identify items that are

disadvantageous to minorities and women. The allegation is that the tests are developed from a perspective that is largely White, middle-class, and male; and that from such a perspective, item content that is socially and educationally biased is likely to be overlooked unless subjected to specific study. Yet, as just indicated, when the total test score is used as the matching variable, the DIF procedures all coerce the finding that if some items have negative DIF, others will have positive DIF; and some writers would maintain that to be even-handed one should scrutinize items with positive DIF for rejection or revision just as one scrutinizes for rejection or revision items with negative DIF. But as just indicated, this practice would fly in the face of the original purpose of DIF analyses, which is to rid the test of any items found to be biased against minorities or women.

One practical way out of this quandary is to examine positive DIF items for any obvious flaws that would clearly disadvantage the White middle-class male and revise only the most egregious among them, but otherwise devote attention only to the items with negative DIF. This proposal will certainly cause some dissent, as will virtually any other proposal. One can only hope that the debate over the dilemma will yield a solution satisfactory both to those who insist on "even-handedness" and those who argue "original purpose".

Yet another way to deal with this matter is to balance the test with different sets of items known to favor different groups; items on Black history and literature, to favor Black students; items relating to persons and personal relationships, art, music, literature, and drama, to favor women; items relating to male sports, war, science, and mathematics, to favor men; and so forth. The natural question here is: What mix of various content areas best reflects reality? And according to whose perception of reality? And would not this procedure open the door to the danger of self-serving efforts to engineer the results of the test? There is no easy answer to these questions, but here, as in many other situations, we must depend on the consensus of committees of educators who are expert in subject-matter and curriculum and closely familiar with the data of group differences. It would be left to these people to design the test content in accordance with current educational philosophies and practices as they understand them.

FURTHER APPLICATIONS OF THE DIFFERENTIAL ITEM FUNCTIONING TECHNOLOGY

Out of the methodological efforts of the last 25 years has emerged an arsenal of methods for the analysis and identification of items with DIF characteristics. An overview at this point suggests that the methods of choice are the Mantel–Haenszel procedure, which permits the examination of differential item difficulty effects, and the comprehensive three-parameter item response theory approach, which permits the observation of differences in any of the characteristics of the

IRT function. A number of questions remain, some related to the specifics of technique, others to broader, philosophical issues. It is expected that continued work on methodology and application will answer some of the present questions, and, more than likely, raise new ones.

In the meantime, we do have a respectable methodology to use in answering some pressing issues of item bias. It should not go unnoticed, however, that the same methodology is useful in other contexts, not necessarily having to do with social bias as such, and it may be useful to identify some of them and urge the continued use of the DIF methodology in those contexts and its introduction into new contexts. Examples of some of these other contexts follow:

During the early 1960s, new types of curricular organization and content were introduced into American secondary school curricula, especially in the teaching of science. For a time, some students were still being taught by means of the traditional textbooks, while others were being introduced to textbooks written to accompany the new curricula. This outcome directly affected the development of the College Board Achievement Tests in science. It was clear that these tests would have to accommodate both the students who had been enrolled in the new curricula and those who had been enrolled in the traditional curricula, and DIF studies were carried out to determine whether there was an equal balance of items of both types (Angoff, 1971).

As mentioned earlier, DIF studies have been used to identify differential response patterns on the Test of English as a Foreign Language for examinees coming from different countries and speaking different languages (e.g., Alderman & Holland, 1981; Angoff & Sharon, 1974). Presumably, information provided by studies of this sort could be valuable for students of linguistics, not to identify biased items but simply to investigate the effect of non-English linguistic differences on English expression.

In the equating of scores on different test forms it is often necessary to use a set of common items to be administered to both groups, each taking a different form. To verify that these common items are in fact common psychologically, having the same meaning (although not necessarily the same difficulty) for both groups, it is found useful to carry out a DIF study. Such studies have in fact revealed important differences (after matching) in the difficulty of these common items, especially when they have been used for equating different forms of an achievement test administered to different groups at different times in the school year (Cook & Petersen, 1987).

As already cited, Gulliksen (1964) carried out some cross-cultural attitude studies, using a procedure similar to the delta-plot (TID) method. With the improved DIF methodology now available, such studies, perhaps using a different procedure, could be carried out on a continuing basis.

On one occasion there was some concern that a for-profit "performance contracting" (as it was then called) organization, engaged to develop an educational program in reading and mathematics and to teach the courses on a trial basis to

children in the elementary schools of a Southern school system, had actually been teaching the children the correct answers to items on the tests that were to be used later for evaluating the program. A DIF study, in which the children's responses on the tests were compared with the response of a control group of children not participating in the special program, confirmed the suspicion that the tests had indeed been compromised; the study made it clear that several items on the tests were inordinately easier, relative to the other items in the tests, for the group who were suspected of having been "taught the test."

Studies of language translations have been carried out by Angoff and Modu (1973), Angoff and Cook (1988), and Hulin, Drasgow, and Parsons (1983), and validated by means of DIF procedures to determine whether an item has the same meaning when expressed in two different languages.

In the mid-1970s, when interest in the SAT score decline was at its peak, it was hypothesized that some of the old items in the common-item sections used for equating the current forms may have changed in difficulty over time and may have affected the score scale in such a way as to cause an artifactual decline in scores. A DIF analysis demonstrated that this was not the case.

It has often been said that the intellectual behaviors of two children of the same mental age but different chronological ages are different—that, for example, a 6-year-old with a mental age of 8 and an 8-year-old with a mental age of 8 are different intellectually. There is not an extensive empirical literature, however, on precisely what the nature of these differences may be. Where do these differences appear and in what particular cognitive dimensions? In the context of test performance, how can they be recognized in the children's responses to test items? If they do not appear in the correct options, do they appear in the distractors; and if not there, then in what respects do these differences exist? Quite possibly, differential item functioning techniques might be of considerable assistance in research of this sort.

Thus, even where item bias as such is not the issue, it is interesting to study the behavior of items and its variation across ethnic, cultural, and linguistic groups and across groups that differ as a function of nationality, geography, time, sex, age, and educational disability. With techniques currently available, one can carry out more rigorous studies of general and specific retardation with possibly useful application in areas of educational and psychological diagnosis. It is maintained that the methods now available for examining differential item functioning can be used for studying a much broader range of problems than the problem of item bias and it is important that we investigate appropriate applications for them.

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